

A Faster Attack on LFSR Recurrences
An Alternative to Berlekamp-Massey for Complete Ciphertexts

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Shenandoah Undergraduate Mathematics Symposium
November 1, 2025

Overview

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- As it turns out, it repeats itself after 31 terms.

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(01111, $x_{n+5} \equiv x_n + x_{n+2}$) $\implies$ 1001101001000010101110110001111

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- Start with two cases $n = 1$ and $n = 2$, then $x_1 = 0, x_2 = 1, x_3 = 1, x_4 = 0$.

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The solution to the system is $c_0 = 1$, $c_1 = 1$, thus we assume that the recurrence is $x_{n+2} \equiv x_n + x_{n+1}$, which **doesn't** fit the sixth digit, so make a new guess x_{n+3} .

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- The Gaussian Elimination Approach is computationally expensive.

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Let N be the length of the shortest recurrence, and M_N be the matrix containing the systems of equations for $[c_1, c_2, \dots, c_n]^\top$, then $\det(M_N) \equiv 1 \pmod{2}$, and for any $n > N$, $\det(M_n) \equiv 0 \pmod{2}$ ¹.

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- 2 Compute its determinant.

$$1001010000\underline{1}00000000000000000000 \quad (3)$$

The first matrix with a determinant of one, with ε many zero determinants after the one, **is** the coefficient matrix for the system of equations, equivalent to the recurrence polynomial.

Main Results

Theorem (Tarter-Herasymov-Priebracha Algorithm)

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Once the bit recurrence string is obtained, recover the recurrence polynomial with BM.

Proof

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- Since $|\gamma|$ divides n by definition, our algorithm will identify it as a potential k . Since it is also a recurrence, it can be reconstructed into β by concatenating it with itself $\frac{n}{|\gamma|}$ times, and thus is contained in θ .

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- We want to show that $\gamma \in \theta$ and we want to show that there does not exist something smaller than γ in θ .
- Since $|\gamma|$ divides n by definition, our algorithm will identify it as a potential k . Since it is also a recurrence, it can be reconstructed into β by concatenating it with itself $\frac{n}{|\gamma|}$ times, and thus is contained in θ .
- To show that there is not a smaller element in θ than γ , we assume there exists $\eta \in \theta$ with $|\eta| < |\gamma|$. But if $\eta \in \theta$, then η can be reconstructed as a recurrence of β by concatenating itself $\frac{n}{|\eta|}$ times, and is therefore a smaller recurrence than γ , which we assumed to be the minimal recurrence. Therefore, there does not exist such an η , and $\min \theta = \gamma$.

Computational Complexity

Time Complexity

Algorithm	Complexity
THP	$O(n + \log(n))$
BM	$O(n^2)$

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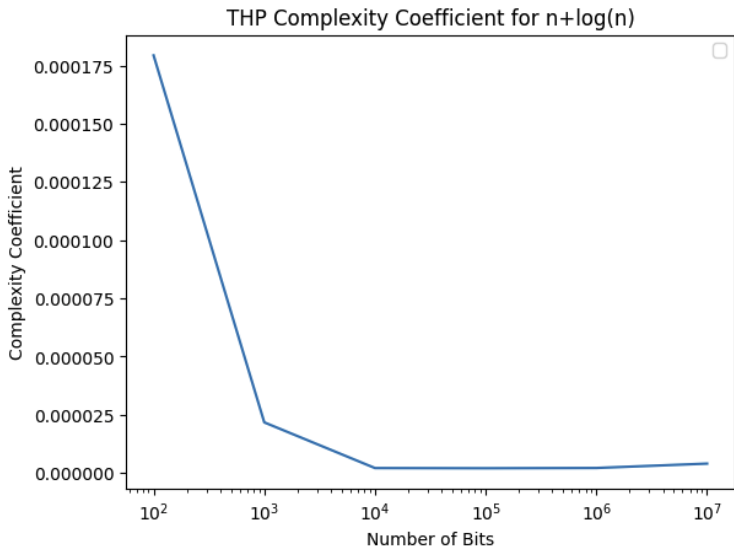
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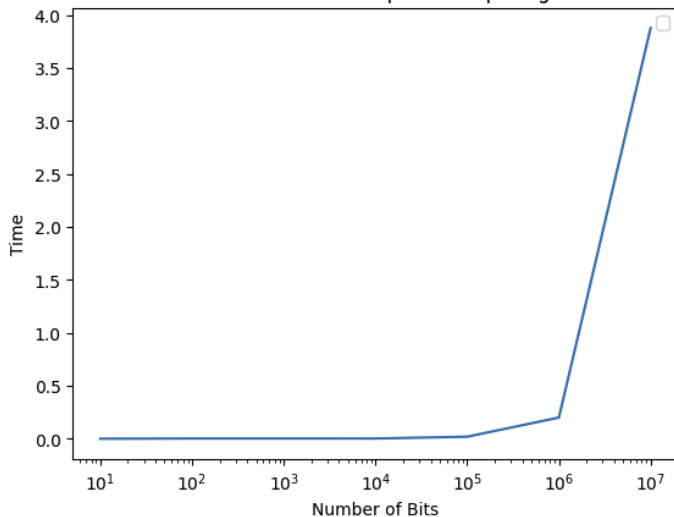
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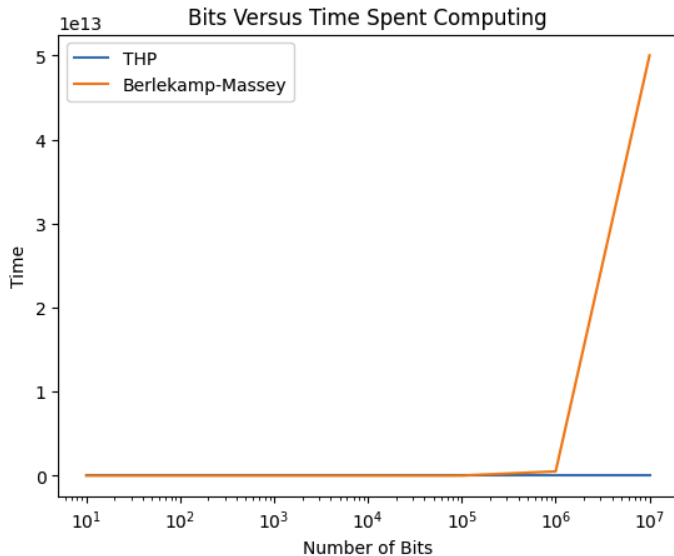
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- 1 THP requires a full string whereas Berlekamp-Massey works for partial string as well.
- 2 Our time complexity prevails for large sequences, unfeasible for Berlekamp-Massey.



Bits Versus Time Spent Computing







GitHub Repository



W. Trappe and L. C. Washington, *Introduction to Cryptography with Coding Theory*, 2nd ed., Pearson Education, 2006.



M. Tomlinson, C. J. Tjhai, M. A. Ambroze, M. Ahmed, M. Jibril, *Error-Correction Coding and Decoding: Bounds, Codes, Decoders, Analysis and Applications*, Springer International Publishing, 2017.



S. Nikolaev, *Advanced Berlekamp–Massey Method as a Basis for Finding Periodicities in Bitstreams*, Cybernetics and Systems Analysis, 61, pp. 3–10, 2025



S. D. Galbraith, *Cryptography and Coding: 11th IMA International Conference*, Cirencester, UK, December 18–20, 2007, Proceedings, Springer, pp. 220–232, 2007



A. Kreczmar and G. Mirkowska, *Mathematical Foundations of Computer Science 1989*, Lecture Notes in Computer Science, vol. 379, Springer, pp. 431–444, 1989.



L. J. Guibas, *Periods in Strings*, Journal Combinatorial Theory, Series A, vol. 30, Issue 1, January 1981, pp. 19-42.



K. Yu, *On binary recurrence sequences*, Indagationes Mathematicae, vol. 6, Issue 3, 1995, pp. 341-354.



Y. Wu, *High-Speed LFSR Decoder Architectures for BCH and GII Codes*, IEEE Journal on Selected Areas in Information Theory, vol. 4, pp. 331-350, 2023